The complete excess-three inverse decoder layer

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For a positive odd integer x, let

$$U_+(x) := \frac{3x+1}{2^{v_2(3x+1)}}. \tag{1}$$

An exponent word $E = (e_1, \ldots, e_h)$ is read from its positive odd start toward its terminal state. We define its mass and excess by

$$\mathrm{B}(E) := e_1 + \cdots + e_h, \qquad t(E) := \mathrm{B}(E) - h. \tag{2}$$

The purpose of this note is to classify every exact inverse word with $t(E) = 3$ at a fixed positive odd terminal state.

We begin by recalling the reverse rule. If x is the current odd terminal state, an exponent-a predecessor is

$$p = \frac{2^a x - 1}{3}. \tag{3}$$

It exists precisely when $3 \nmid x$ and

$$a \equiv \begin{cases} 1 \pmod 2, & x \equiv 2 \pmod 3, \\ 0 \pmod 2, & x \equiv 1 \pmod 3. \end{cases} \tag{4}$$

In the shifted coordinate $X := x + 1$, the four reverse operations used below are

$$X \longmapsto \frac{2X}{3}, \qquad X \longmapsto \frac{4X-2}{3}, \qquad X \longmapsto \frac{8X-6}{3}, \qquad X \longmapsto \frac{16X-14}{3}. \tag{5}$$

Theorem 1 (complete excess-three classification)**.** *Let r be a positive odd integer and put*

$$b := v_3(r+1), \qquad s := \frac{r+1}{3^b}. \tag{6}$$

There are exactly four families of exact predecessor words E with $t(E) = 3$.

Family I: one exponent four. *Define*

$$Q := 2^{b+3}s - 7, \qquad c := v_3(Q). \tag{7}$$

For every $1 \le j \le c$, there is one exact word

$$\boxed{E = 1^{j-1}4\,1^b, \qquad |E| = b + j, \qquad z + 1 = \left(\frac{2}{3}\right)^j Q.} \tag{8}$$

This family is empty when $c = 0$.

Family II: an exponent three followed in reverse by an exponent two. *For every $0 \le k < b$, define*

$$M_k := 2^{k+2}3^{b-k-1}s - 1, \qquad c_k := v_3(M_k), \qquad u_k := \frac{M_k}{3^{c_k}}, \tag{9}$$

We also define

$$P_k := 2^{c_k+2}u_k - 1, \qquad d_k := v_3(P_k). \tag{10}$$

For every $1 \le j \le d_k$, there is one exact word

$$\boxed{E = 1^{j-1}2\,1^{c_k}3\,1^k, \qquad |E| = k + c_k + j + 1, \qquad z + 1 = \left(\frac{2}{3}\right)^j P_k.} \tag{11}$$

The branch indexed by k is empty when $d_k = 0$.

Family III: an exponent two followed in reverse by an exponent three. *Define*

$$N_1 := 2^{b+1}s - 1, \qquad b_1 := v_3(N_1). \tag{12}$$

This family is empty when $b_1 = 0$. If $b_1 \ge 1$, put

$$s_1 := \frac{N_1}{3^{b_1}}. \tag{13}$$

For every $0 \le k < b_1 - 1$, define

$$R_k := 2^{k+3}3^{b_1-k-2}s_1 - 1, \qquad f_k := v_3(R_k). \tag{14}$$

For every $0 \le \ell \le f_k$, there is one exact word

$$\boxed{E = 1^\ell 3\,1^k 2\,1^b, \qquad |E| = b + k + \ell + 2, \qquad z + 1 = 2\left(\frac{2}{3}\right)^\ell R_k.} \tag{15}$$

Family IV: three exponents two. *Suppose $b_1 \geq 1$, and use s_1 from (13). Define*

$$N_2 := 2^{b_1+1}s_1 - 1, \qquad b_2 := v_3(N_2). \tag{16}$$

This family is empty when $b_2 = 0$. If $b_2 \geq 1$, put

$$s_2 := \frac{N_2}{3^{b_2}}, \qquad N_3 := 2^{b_2+1}s_2 - 1, \qquad b_3 := v_3(N_3). \tag{17}$$

For every $1 \leq j \leq b_3$, there is one exact word

$$\boxed{\begin{aligned} E &= 1^{j-1}2\,1^{b_2-1}2\,1^{b_1-1}2\,1^{b}, \\ |E| &= b + b_1 + b_2 + j, \\ z + 1 &= \left(\frac{2}{3}\right)^{j} N_3. \end{aligned}} \tag{18}$$

There are no other exact words of excess three.

For the horizon statement below, we set $b_2 = b_3 := 0$ when $b_1 = 0$, and we set $b_3 := 0$ when $b_1 \geq 1$ and $b_2 = 0$.

Proof. Every exponent contributes $e_i - 1$ to the excess. Therefore, an excess-three word has exactly one of the following forms: one exponent four, one exponent three together with one exponent two, or three exponents two. We read each possibility backward from r.

The shifted terminal state is $3^b s$. Precisely b consecutive exponent-one inverses are initially available, and after all of them the shifted state is $2^b s$.

Suppose first that the word contains one exponent four. This exponent must occur at the first even-parity junction, after all b exponent-one inverses. Reversing it gives

$$\frac{16 \cdot 2^b s - 14}{3} = \frac{2Q}{3}. \tag{19}$$

This is integral exactly when $c \geq 1$. Its 3-adic valuation is $c - 1$, so exactly $j - 1$ additional exponent-one inverses are available for $1 \leq j \leq c$. Their shifted starts are $(2/3)^j Q$. This proves Family I.

Next suppose that an exponent three is encountered first when reading backward. It must occur strictly inside the initial exponent-one run. After k exponent-one inverses, where $0 \leq k < b$, the shifted state is

$$2^k 3^{b-k} s. \tag{20}$$

Reversing exponent three gives $2M_k$. The next non-one inverse must be the unique exponent two, so all c_k available exponent-one inverses must first be

taken. The shifted state at that junction is $2^{c_k+1}u_k$, and the exponent-two inverse gives

$$\frac{4 \cdot 2^{c_k+1}u_k - 2}{3} = \frac{2P_k}{3}. \tag{21}$$

The remaining exponent-one inverses give exactly (11). This proves Family II.

Suppose instead that the exponent two is encountered first. It must occur at the first even-parity junction. Its shifted output is

$$\frac{4 \cdot 2^b s - 2}{3} = \frac{2N_1}{3}, \tag{22}$$

so it exists exactly when $b_1 \geq 1$. If the exponent three is encountered next, it must occur strictly inside the following run of $b_1 - 1$ exponent-one inverses. After k of these inverses the shifted state is

$$2^{k+1}3^{b_1-k-1}s_1. \tag{23}$$

Reversing exponent three gives $2R_k$, after which precisely f_k exponent-one inverses remain available. This proves Family III.

It remains to consider three exponents two. Parity forces each one to occur at the next junction, after the current exponent-one run is exhausted. The first junction is described by (22). After the next $b_1 - 1$ exponent-one inverses, the shifted state is $2^{b_1}s_1$, and the second exponent-two inverse gives $2N_2/3$. After the following $b_2 - 1$ exponent-one inverses, the shifted state is $2^{b_2}s_2$, and the third exponent-two inverse gives $2N_3/3$. Reversing a final $j - 1$ exponent-one steps gives (18).

The three partitions of the excess are exhaustive. In each partition, the parity rule forces every non-one exponent into the positions used above. Therefore, the four displayed families contain every exact excess-three word. □

Corollary 2 (finite excess-three horizon)**.** *With the notation of Theorem 1, define*

$$H_I := b + c, \tag{24}$$

$$H_{II} := \max_{0 \leq k < b}(k + c_k + d_k + 1), \qquad H_{III} := \max_{0 \leq k < b_1 - 1}(b + k + f_k + 2), \tag{25}$$

where the maximum of an empty set is zero, and define

$$H_{IV} := \begin{cases} b + b_1 + b_2 + b_3, & b_1 b_2 b_3 > 0, \\ 0, & b_1 b_2 b_3 = 0. \end{cases} \tag{26}$$

Then no exact h-step predecessor word of r has excess three when

$$\boxed{h > H_3(r) := \max(H_I, H_{II}, H_{III}, H_{IV}).} \tag{27}$$

If $H_{\le 2}(r)$ denotes the horizon from the excess-zero, excess-one, and excess-two classifications, then every exact prefix with

$$h > H_{\le 3}(r) := \max\big(H_{\le 2}(r), H_3(r)\big) \tag{28}$$

has excess at least four.

Proof. The largest word length in each family is the corresponding quantity in (24)–(26). Theorem 1 proves (27). Combining it with the three earlier decoder layers proves (28). □

Corollary 3 (fixed-seed inequalities)**.** *Let n be the globally least positive odd integer with an unbounded accelerated orbit, and suppose that the orbit of r is unbounded. Every branch in Theorem 1 satisfies its corresponding inequality*

$$\boxed{\begin{aligned} Q &\ge (n+1)\left(\frac{3}{2}\right)^{j}, && 1 \le j \le c, \\ P_k &\ge (n+1)\left(\frac{3}{2}\right)^{j}, && 1 \le j \le d_k, \\ R_k &\ge \frac{n+1}{2}\left(\frac{3}{2}\right)^{\ell}, && 0 \le \ell \le f_k, \\ N_3 &\ge (n+1)\left(\frac{3}{2}\right)^{j}, && 1 \le j \le b_3. \end{aligned}} \tag{29}$$

In particular, every minimum-channel prefix of excess at most three is now contained in a finite explicit valuation ladder. Beyond $H_{\le 3}(r)$, that channel has excess at least four.

Proof. Every positive predecessor of r also has an unbounded forward orbit. Its start is therefore at least n. Substitution of the four shifted starts in (8), (11), (15), and (18) proves (29). The final assertion is Corollary 2. □

The accompanying exact certificate is

`verify_divergence_record_excess_three.py`.

It independently enumerates every reverse word of excess three and checks the four families for all 50,001 odd terminal states $r \le 100{,}001$ using exact integer arithmetic.